

Descriptor-surrogate attribution across 13 Matbench tasks

Within-task normalized mean absolute SHAP values (%)

		Surrogate fidelity																	
		CV	R ²																
Experimental metallicity	0.642 Moderate	20.0	18.9	8.3	6.6	0.4	0.4	2.7	1.6	5.4	0.0	5.1	1.3	4.0	0.3	4.0	21.0	100.0	
Glass-forming ability	0.327 Low	6.3	7.6	11.9	16.0	1.2	0.6	1.6	4.5	0.9	0.0	1.2	3.1	3.0	1.9	2.3	37.9	100.0	
MP metallicity	0.544 Moderate	22.7	17.6	4.0	2.4	2.5	4.2	3.1	1.5	6.3	1.5	5.7	0.1	4.1	0.7	2.1	21.5	100.0	
Refractive index	0.616 Moderate	1.4	1.5	0.8	0.7	8.2	3.1	3.3	0.8	1.6	13.4	1.7	4.1	1.1	0.3	3.0	55.0	100.0	
Experimental band gap	0.379 Low	13.7	13.1	4.8	1.2	3.9	0.4	4.2	4.3	6.7	0.0	5.5	0.5	5.5	1.7	3.7	30.8	100.0	
2D exfoliation energy	0.552 Moderate	0.8	1.1	0.2	0.7	2.3	3.6	2.6	1.7	0.8	3.9	0.8	1.1	0.5	2.7	2.1	75.1	100.0	
Shear modulus	0.739 Moderate	34.5	26.2	1.7	2.6	0.1	7.5	0.6	0.6	2.1	2.5	2.1	0.7	0.8	0.6	0.3	17.1	100.0	
Bulk modulus	0.801 High	32.1	23.6	4.0	4.9	2.0	6.0	1.1	1.0	1.3	3.3	1.1	0.3	0.7	0.0	0.3	18.3	100.0	
MP formation energy	0.833 High	13.5	8.3	11.9	11.6	8.8	1.1	2.9	1.4	1.6	2.0	1.2	0.0	0.5	0.0	2.9	32.3	100.0	
MP band gap	0.112 Low	1.7	2.5	0.5	3.0	6.0	4.1	5.7	9.0	0.2	1.8	0.7	3.0	0.3	3.8	1.1	56.6	100.0	
Perovskite formation energy	0.560 Moderate	12.5	8.4	3.0	3.6	2.1	3.7	4.3	3.7	1.2	0.3	2.9	0.1	1.7	1.4	1.2	49.9	100.0	
Optical phonon frequency	0.723 Moderate	6.2	0.5	12.5	11.8	2.8	8.6	0.1	0.4	4.2	0.3	4.6	0.7	4.1	2.9	3.0	37.3	100.0	
Steel yield strength	0.433 Low	1.6	1.1	0.1	1.2	11.8	3.9	9.5	4.8	0.3	0.0	0.3	12.2	0.2	12.8	9.3	30.9	100.0	
		Mean atomic number	Mean atomic mass	Atomic-number dispersion	Atomic-mass dispersion	Electronegativity dispersion	Mean periodic-table period	Mean atomic radius	Period dispersion	Maximum atomic number	Density	Maximum atomic mass	Metal fraction	Atomic-number range	Group-number dispersion	Mean electronegativity	Other descriptors	Row total	